



SAPIENZA
UNIVERSITÀ DI ROMA

Py2DIC: Python software for 2D Digital Image Correlation (DIC)



py2DIC

OPEN SOURCE DIGITAL IMAGE CORRELATION

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1 Py2DIC software

Py2DIC is a free, open-source software developed at the Geodesy and Geomatics Division of Sapienza University of Rome for 2D Digital Image Correlation (DIC). It is based on the template matching method and convolution algorithms to compute 2D displacement and strain fields. The software compares a series of images of a planar surface collected at different stages of deformation and tracks the movement of pixels inside the Area of Interest (AOI) using template matching algorithms (Figure 1.1).

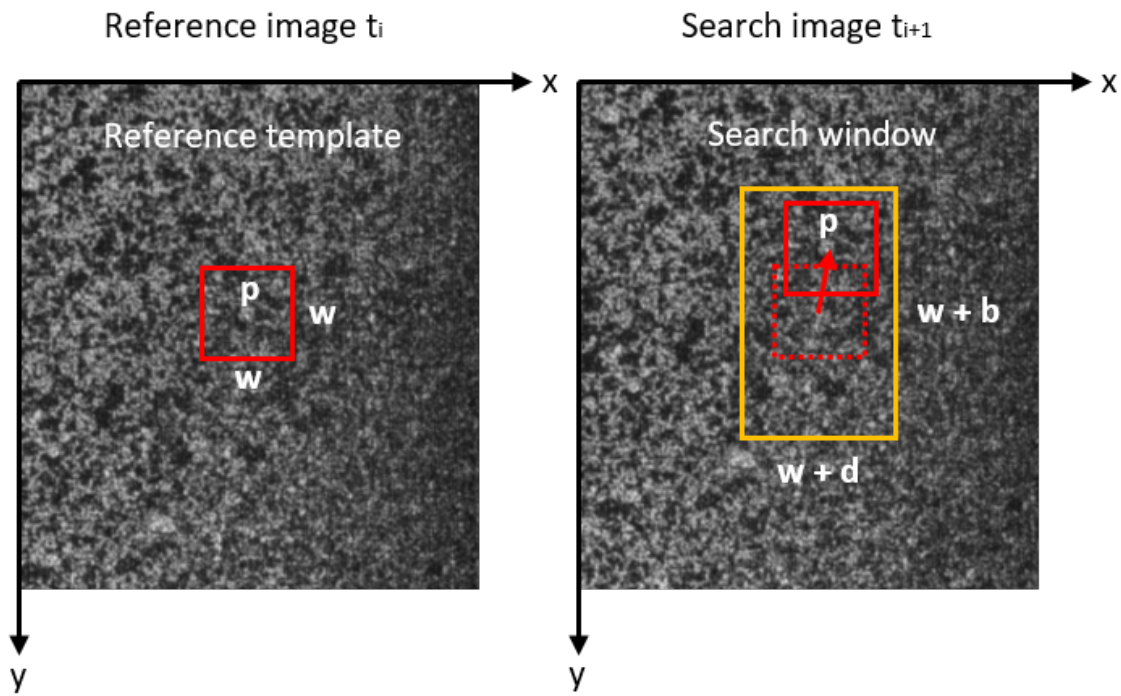


Figure 1.1: Template matching [1].

Once the displacements are computed, a Gaussian filter is applied to smooth the displacements estimated at each pixel and reduce noise. The smoothed displacements are then differentiated using the centred difference approximation, for which the weights (up to the eighth order of accuracy) are generated following the approach described in [5]. Finally, the Green Lagrangian strains are computed. For details on the calculation of displacement and strain fields, refer to [3]. Additional information and testing of the software are provided in [2, 1, 6].

The software is written in Python, and the source code is freely available at <http://github.com/Geod-Geom/py2DIC>. Py2DIC is provided with a Graphical User Interface (GUI) and leverages the potentialities of OpenCV [4] (Figure 1.2).

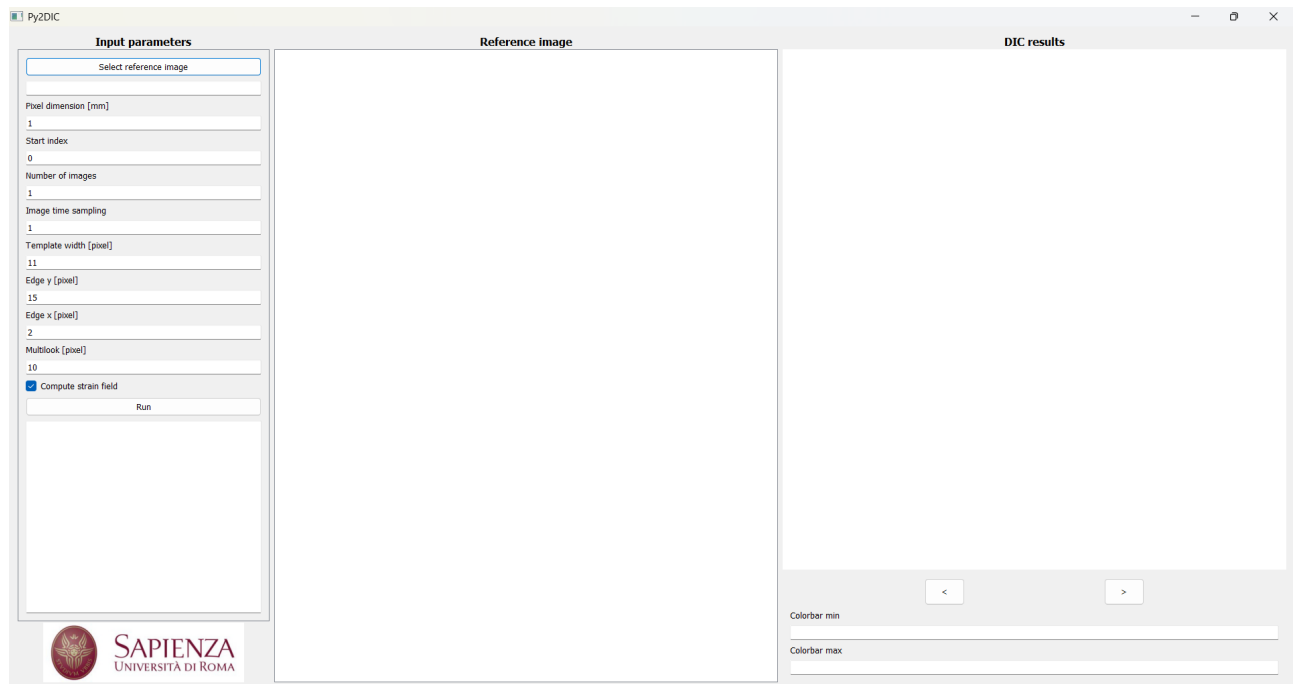


Figure 1.2: Py2DIC GUI.

At the end of the processing, the software returns the displacement and strain fields (Green Lagrangian strains). An example of the results is shown in Figure 1.3.

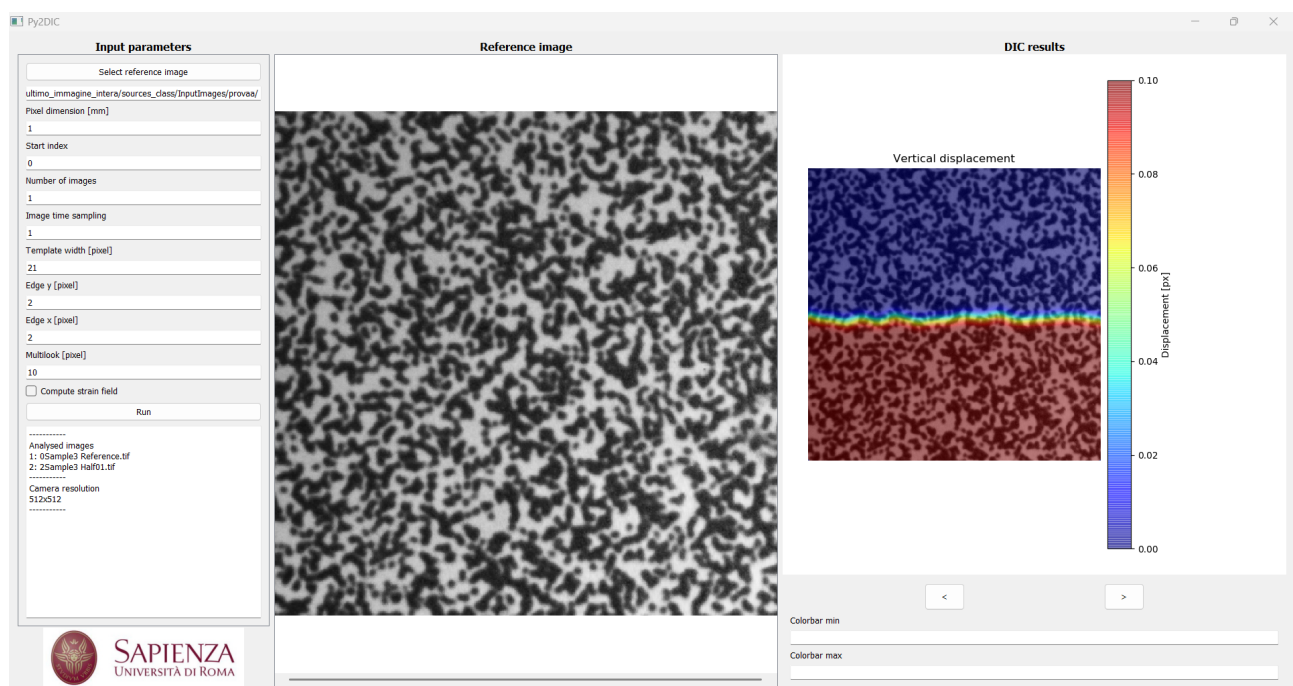


Figure 1.3: Py2DIC outcomes.

2 Installation

- Install Anaconda following the instructions available at <https://docs.anaconda.com/anaconda/install/>.
- Open the Anaconda Prompt.

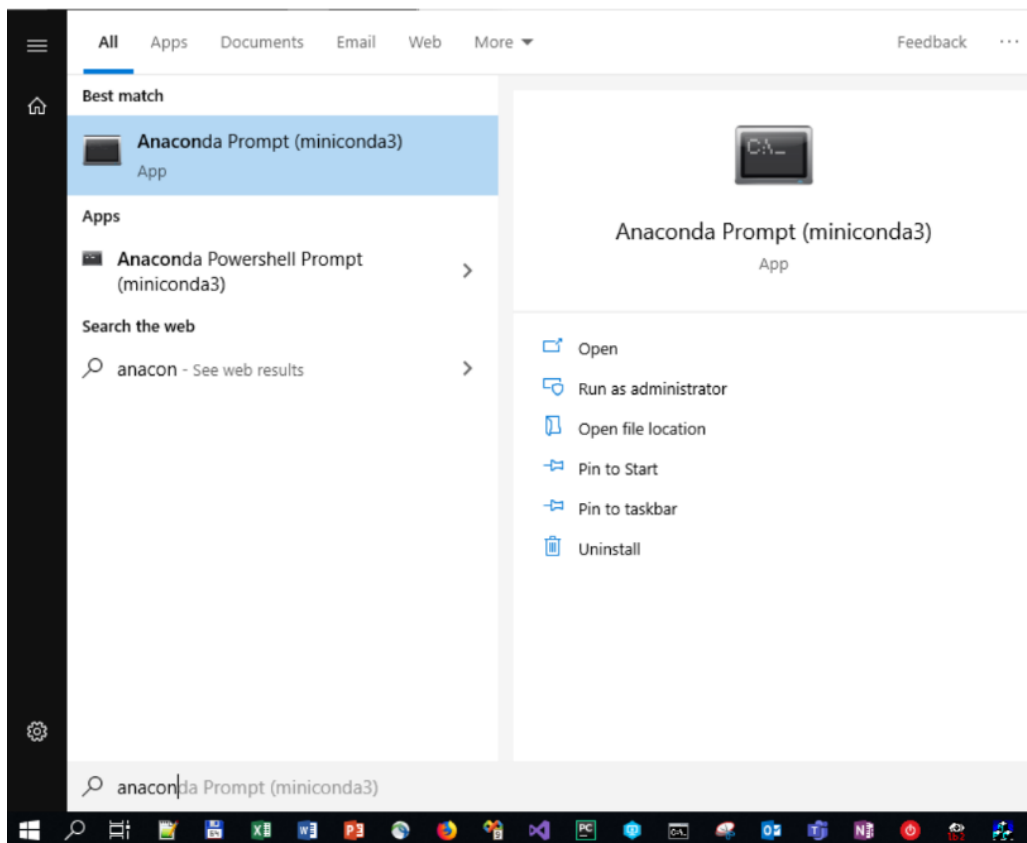


Figure 2.1: Open the Anaconda Prompt.

- Create an environment named py2DIC in the Anaconda Prompt using the command:
conda create --name py2DIC.

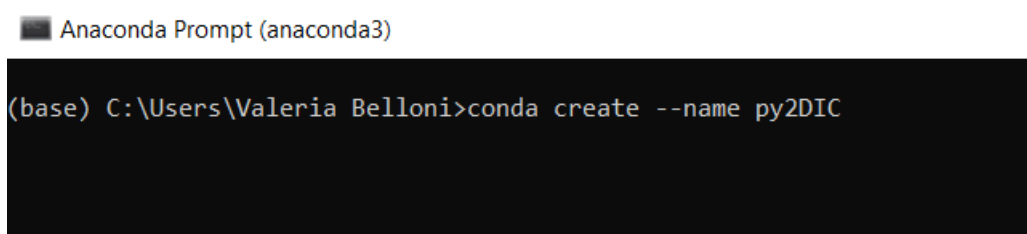


Figure 2.2: Create the environment in the Anaconda Prompt.

- Activate the environment in the Anaconda Prompt using the command:
conda activate py2DIC.

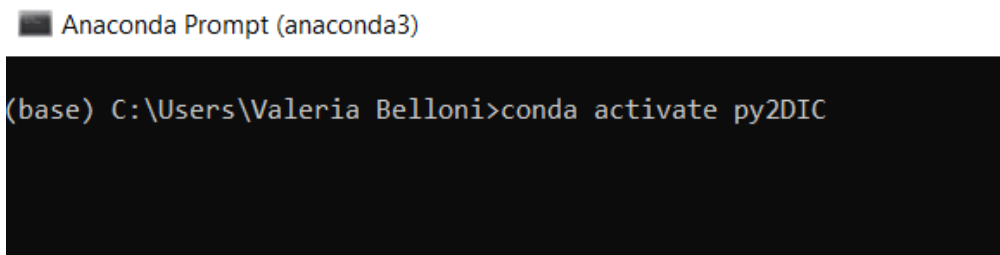


Figure 2.3: Activate the environment in the Anaconda Prompt.

- Install all the dependencies in the Anaconda Prompt using the following command:
conda install -c conda-forge opencv matplotlib scipy joblib astropy pyqt

3 User guide

3.1 Getting started

To download the software:

- Go to the GitHub repository <https://github.com/Geod-Geom/py2DIC>.
- Click the green *Code* button.
- From the dropdown menu, select *Download ZIP*.
- Save the ZIP file to your desired location on your computer.
- Extract the contents of the ZIP file to access the software.

3.1.1 Run via batch file

- Open the batch file available in the folder *run_py2DIC.bat*.
- Update the paths highlighted in Figure 3.1 and save the file. The first path refers to the conda environment; the second is the path to the folder downloaded from GitHub.
- Double-click the batch file (run_py2DIC.bat) to launch the GUI (Figure 3.2).

```

1  @echo OFF
2  rem How to run a Python script in a given conda environment from a batch file.
3
4  rem It doesn't require:
5  rem - conda to be in the PATH
6  rem - cmd.exe to be initialized with conda init
7
8  rem Define here the path to your conda installation
9  set "CONDAPATH=C:\Users\Public\anaconda3\"
10 rem Define here the name of the environment
11 set ENVNAME=py2DIC
12
13 rem The following command activates the base environment.
14 rem call C:\ProgramData\Miniconda3\Scripts\activate.bat C:\ProgramData\Miniconda3
15 if %ENVNAME%==base (set ENVPATH=%CONDAPATH%) else (set ENVPATH=%CONDAPATH%\envs\%ENVNAME%)
16
17 rem Activate the conda environment
18 rem Using call is required here, see: https://stackoverflow.com/questions/24678144/conda-environments-and-bat-files
19 call %CONDAPATH%\Scripts\activate.bat %ENVPATH%
20
21 rem Run a python script in that environment
22 REM Start the FEED2 GUI
23 python.exe "C:\Users\py2DIC-master\sources\GUI.py"
24 IF %ERRORLEVEL% NEQ 0 PAUSE
25 rem Deactivate the environment
26 call conda deactivate
27
28 rem If conda is directly available from the command line then the following code works.
29 rem call activate someenv
30 rem python script.py
31 rem conda deactivate
32
33 rem One could also use the conda run command
34 rem conda run -n someenv python script.py

```

Figure 3.1: Batch file.

3.1.2 Run via Anaconda prompt

- Open the Anaconda Prompt and activate the py2DIC environment:
conda activate py2DIC
 - Navigate to the path of the Py2DIC folder using the command:
cd "path to GUI.py"
 - Use the following command to activate ipython:
ipython
 - Use the following command to launch the GUI (Figure [3.2](#)):
run GUI.py
-

3.2 Py2DIC settings

The GUI enables the user to set the input parameters (left panel), visualise the selected reference image and define the AOI (centre panel) and inspect the results (right panel) (see Figure 3.2).

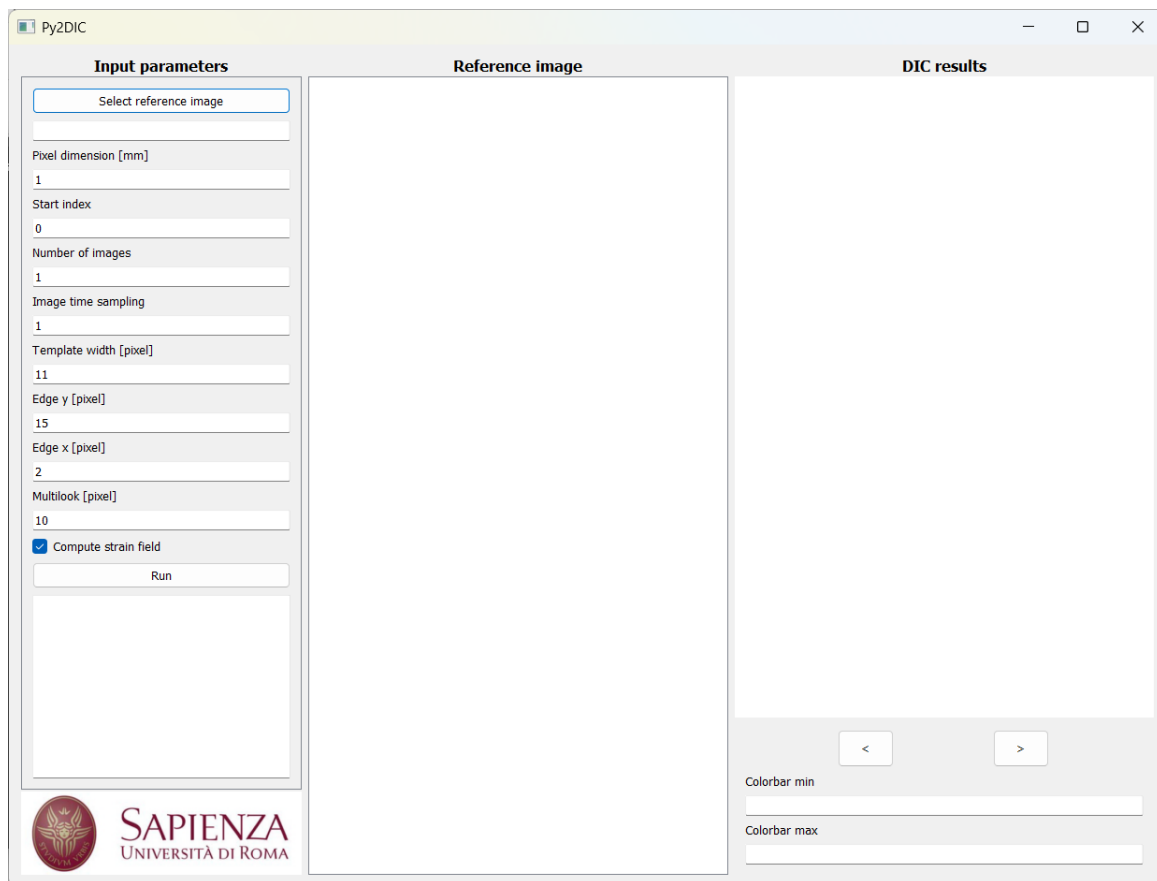


Figure 3.2: Graphical User Interface.

Left panel:

- Select the reference image in the folder using the available button. The image then appears in the central panel.
- Define the pixel dimension to convert the results from pixels to mm. If not available, keep the default value 1. In this case, the results are provided in pixel units.
- Define the start index, which refers to the first image. If set to 0, the reference image is the first image in the folder.
- Define the number of images. Set to 1 to process a single image pair. To process more than one image incrementally, increase this parameter.

- Define the image time sampling, which enables the selection of the deformed image. If the start index is 0, the number of images is 1, and the image time sampling is set to 1 the deformed image is the second image in the folder. If the start index is 0, the number of images is set to 1 and the image time sampling is set to 2, the deformed image is the third image in the folder.
- Define the dimension of the template.
- Define the edges that should be larger than the maximum expected displacements.
- Define the multilook to reach the sub-pixel level. The default value is 10.
- Decide if you want to compute the strains.
- Once the parameters are set, run the program using the *Run* button.

Central panel:

- Visualise the reference image.
- Select the AOI by clicking two points that define the upper left and bottom right corners of a rectangle (Figure 3.3). If the rectangle is not defined, the entire image is processed.

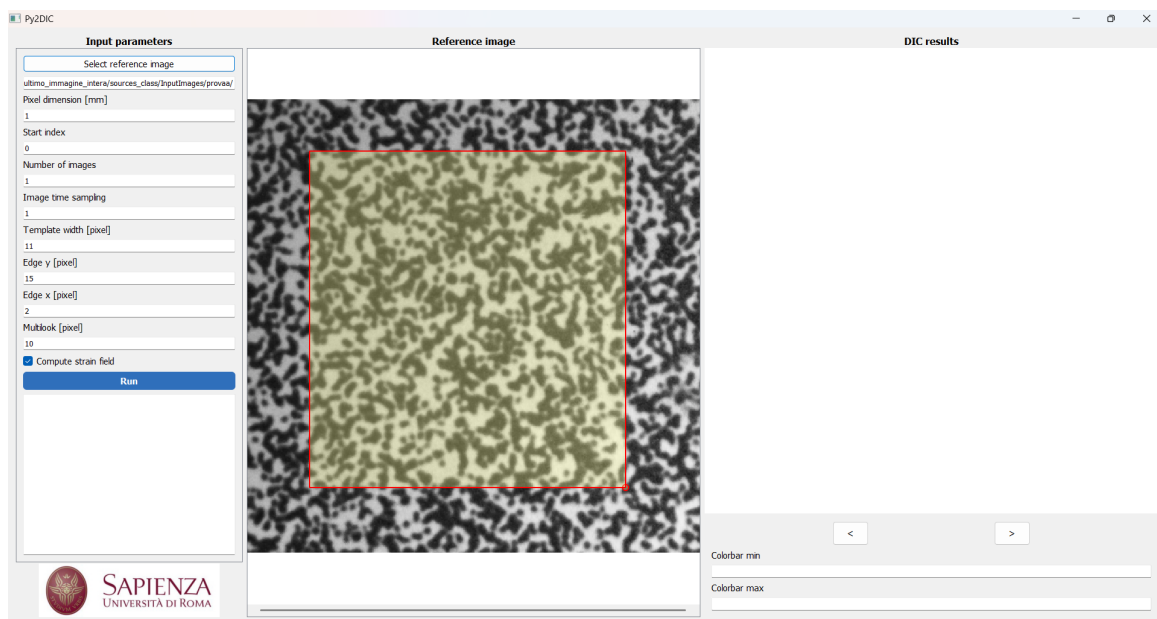


Figure 3.3: AOI selection.

Right panel:

- Push arrow buttons to visualise all the plots.
- Set colorbar min and max values to change the colorbar limits.

4 Testing with samples from the DIC Challenge

The sets of input images available to test the software are part of the Society for Experimental Mechanics (SEM) 2D DIC challenge datasets <https://www.idics.org/challenge/>. The DIC challenge offers a set of common image datasets that can be used to validate and enhance commercial and academic DIC software solutions. A detailed explanation of the dataset, processing, and results of the DIC challenge can be found at <https://link.springer.com/article/10.1007/s11340-017-0349-0>.

4.1 Sample 16

Sample 16 is a rigid-body experimental translation dataset. To process Sample 16 data, download the Sample16.zip folder at <https://drive.google.com/drive/folders/1tNUKPJ7UJ0m23JhERtkrIy5gSBiwV3Dj>. Select the reference image (50pix-test1-0000_0-Dec-Avg 00000.tif) and the deformed image (50pix-test1-0050_0-Dec-Avg 00050.tif) from the Sample16.zip folder. Copy the two images into a new folder. Open the GUI and select the reference image. Set the following parameters and keep the other parameters by default:

- Template = 11 pixel.
- Edges = 2 pixel.
- Multilook = 100.
- Do not computer strain.

The expected results are shown in Figure 4.1. For a proper comparison, set the min and max values of the colorbar as suggested in the figure.

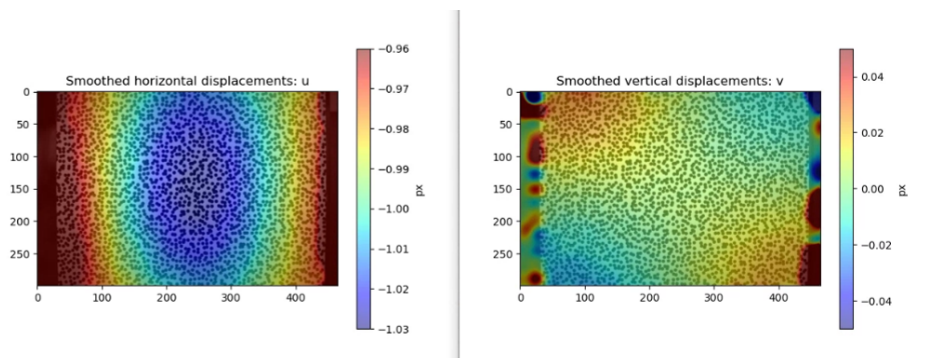


Figure 4.1: Py2DIC outcomes.

4.2 Sample 12

Sample 12 is an experimental set of 12 images (resolution 400×1040 pixels) of a steel plate with a hole in the middle being loaded in tension. Regarding the specimen, a painted speckle pattern was used during the tensile test. As this is an experimental image series, there is no ground truth data, but it is useful for round-robin type tests where different codes for DIC. To process Sample 16 data, download the Sample12.zip folder at <https://drive.google.com/drive/folders/1tNUKPJ7UJ0m23JhERtkrIy5gSBIwV3Dj>. Select the reference image (ohtcfrp_00.png) and the deformed image (ohtcfrp_11.png) and copy the images into a new folder. Keep everything by default and run the program.

The expected results are shown in Figure 4.2. The expected results are shown in Figure 4.1. For a proper comparison, set the min and max values of the colorbar as suggested in the figure.

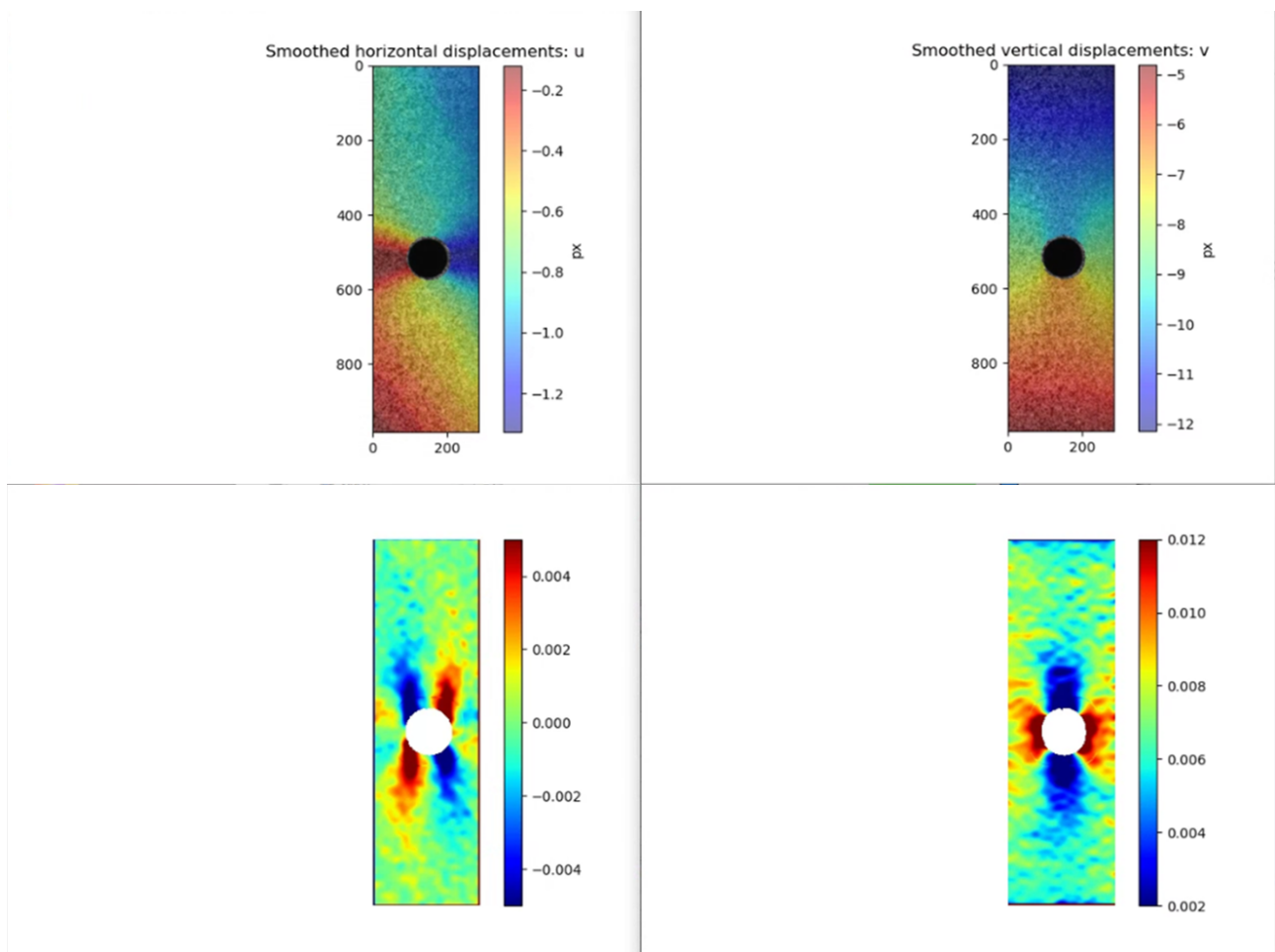


Figure 4.2: Py2DIC outcomes.

4.3 Sample 3b

Sample 3b and Sample 3 are examples of good contrast and low noise images shifted using the FFT function. Sample 3 is a rigid body shift of 0.1 pixels per image and will nicely reveal both the DIC interpolation bias and variance. Sample 3b (resolution of 500×500 pixels) uses the same FFT shift; however, only the bottom half of the image is shifted. Sample 3b can be used to look at the effect of discontinuities in the results [7]. To process Sample 3b data, download the Sample16.zip folder at <https://drive.google.com/drive/folders/1tNUKJP7UJ0m23JhERtkrIy5gSBIwV3Dj>. Select the reference image (0Sample3 Reference.tif) and the deformed image (Sample3 Half01.tif). Rename the image Sample3 Half01.tif as 2Sample3 Half01.tif and copy the images into a new folder. Set the following parameters and keep the other parameters by default:

- Template = 21 pixel.
- Edges = 2 pixel.
- Multilook = 10.
- Do not computer strain.

The expected results are shown in Figures 4.3 and 4.4.

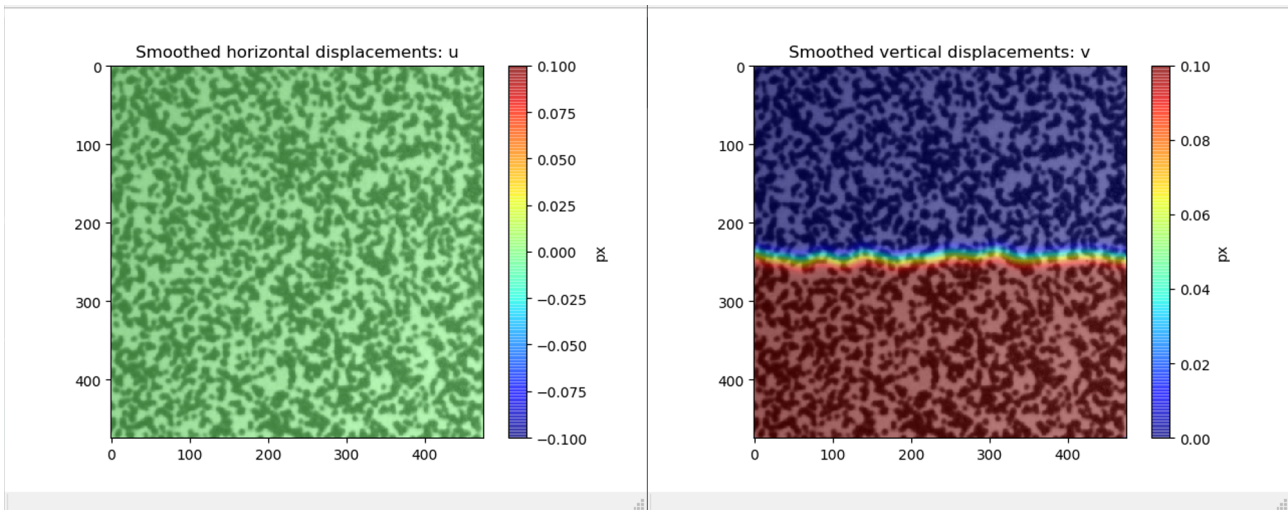


Figure 4.3: Py2DIC outcomes.

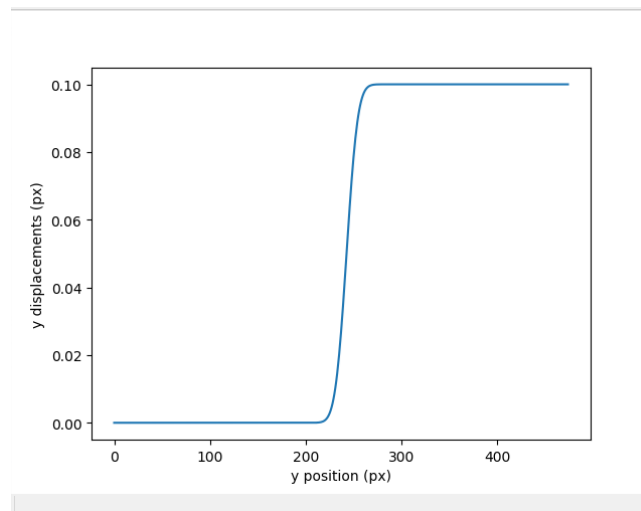


Figure 4.4: Py2DIC outcomes.

Bibliography

- [1] Valeria Belloni et al. *Crack monitoring of masonry walls with standard and enhanced Digital Image Correlation methods*. Tech. rep. 2025. 8 pp. DOI: [10.5445/IR/1000180531](https://doi.org/10.5445/IR/1000180531).
 - [2] Valeria Belloni et al. ‘Digital Image Correlation from commercial to FOS software: a mature technique for full-field displacement measurements’. In: *The International Archives of the Photogrammetry, Remote Sensing and Spatial Information Sciences* XLII-2 (2018), pp. 91–95. URL: <https://isprs-archives.copernicus.org/articles/XLII-2/91/2018/>.
 - [3] Valeria Belloni et al. ‘py2DIC: A New Free and Open Source Software for Displacement and Strain Measurements in the Field of Experimental Mechanics’. In: *Sensors* 19.18 (2019). ISSN: 1424-8220. URL: <https://www.mdpi.com/1424-8220/19/18/3832>.
 - [4] Gary Bradski and Adrian Kaehler. *Learning OpenCV: Computer vision with the OpenCV library*. ” O’Reilly Media, Inc.”, 2008.
 - [5] Bengt Fornberg. ‘Generation of finite difference formulas on arbitrarily spaced grids’. In: *Mathematics of computation* 51.184 (1988), pp. 699–706.
 - [6] Roberta Ravanelli et al. ‘A new Digital Image Correlation software for displacements field measurement in structural applications’. In: *The International Archives of the Photogrammetry, Remote Sensing and Spatial Information Sciences* XLII-4/W2 (2017), pp. 139–145. URL: <https://isprs-archives.copernicus.org/articles/XLII-4-W2/139/2017/>.
 - [7] Philip Reu et al. ‘DIC Challenge 2.0: Developing Images and Guidelines for Evaluating Accuracy and Resolution of 2D Analyses: Focus on the Metrological Efficiency Indicator’. en. In: (2022-01-04 05:01:00 2022). URL: https://tsapps.nist.gov/publication/get_pdf.cfm?pub_id=932767.
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